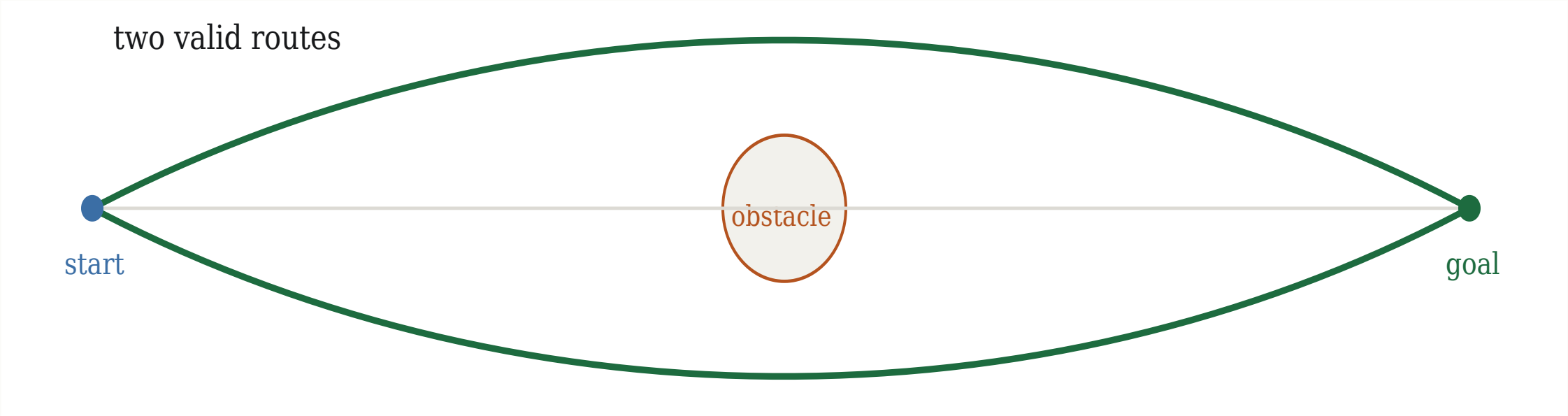
MOTIVATION

Diffusion and Diffusion Policy

From noisy vectors to robot action sequences



> One observation. Several successful action sequences.

start · candidate routes · obstacle

Lecture 11 · November 25, 2026

Mathematics for Robotics · ROB-GY 6013 · NYU Tandon · Jing Zhang · Fall 2026

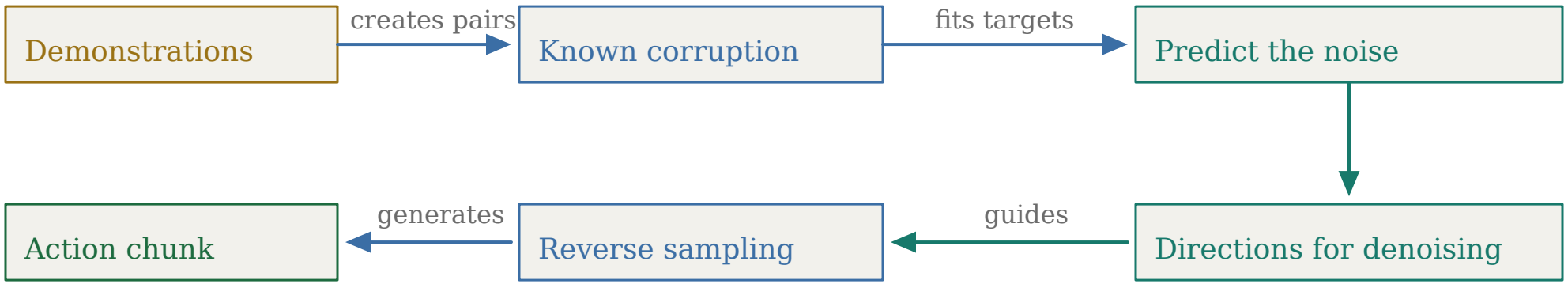
Diffusion Policy §4.1 · Chi et al. (2023/2024)

ROADMAP

Diffusion and Diffusion Policy

A route from demonstrations to actions

Demonstrations supply training data. Repeated denoising generates an action sequence.



DDPM §3; Diffusion Policy §2.3 · Ho, Jain & Abbeel (2020) · Chi et al. (2023/2024)

RECAP

Diffusion and Diffusion Policy

Four familiar tools meet one new task

Recall: Gaussian mean/covariance, input gradients, squared error, iterative updates.

$$\mathbb{E}\|Y - g(X)\|^2$$

- X, Y random input and target
- $g(X)$ prediction of Y from X
- $\mathbb{E}$ average over input–target pairs

Gaussian noise

Input gradient

Least squares

Iterative update

DL §3.8; DDPM §3.2 · Ho, Jain & Abbeel (2020) · Goodfellow et al. (2016)

Multimodal actions

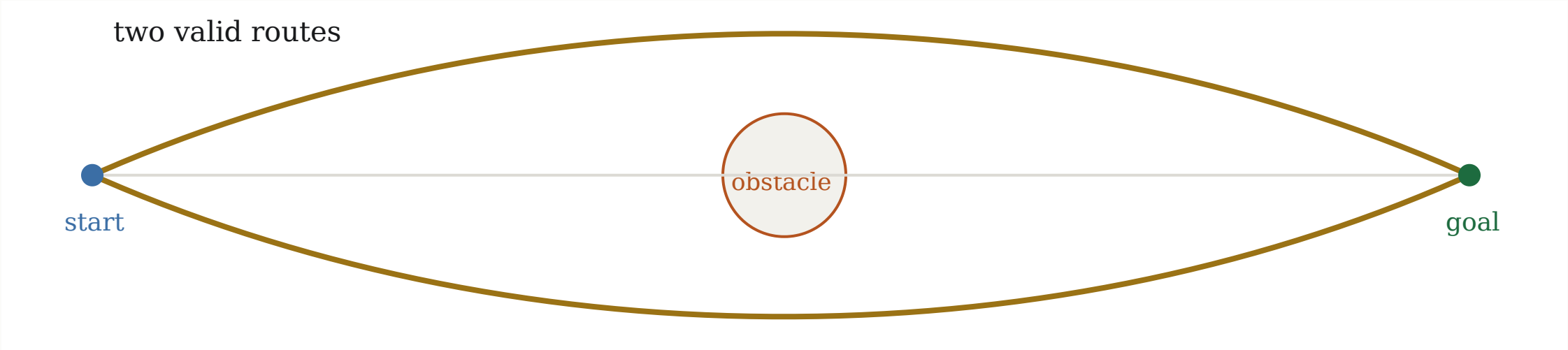
Two valid actions can have an invalid mean

A multimodal distribution has several separated peaks in its probability mass or density.

$$P(A = -2 \mid o) = P(A = 2 \mid o) = \tfrac{1}{2}$$

A scalar route choice; -2 left, $+2$ right

o fixed obstacle observation



Diffusion Policy §4.1 · Chi et al. (2023/2024)

DERIVATION

Multimodal actions

Squared error selects a conditional mean

Squared-error risk is the expected squared prediction error; assume a finite second moment.

$$\mathbb{E}[(A-u)^2 \mid o] = \operatorname{Var}(A \mid o) + (u - \mathbb{E}[A \mid o])^2$$

- u chosen scalar prediction
- A, o action and observation

DL §3.8; course derivation · Goodfellow et al. (2016)

Multimodal actions

Squared error selects a conditional mean

Expand around the conditional mean

Fix o. Let $m = \mathbb{E}[A \mid o]$; u is a scalar prediction. Assume a finite second moment.

1. $A - u = (A - m) + (m - u)$.
2. $(A - u)^2 = (A - m)^2 + 2(A - m)(m - u) + (m - u)^2$.
3. $\mathbb{E}[A - m \mid o] = 0$, so the averaged cross term is zero.

DL §3.8; course derivation · Goodfellow et al. (2016)

Multimodal actions

Squared error selects a conditional mean

Only one term depends on the prediction

m is fixed when u changes.

1. $\mathbb{E}[(A-u)^2 \mid o] = \mathrm{Var}(A \mid o) + (m-u)^2$.
2. The variance is independent of u; the remaining square is minimized at zero.
3. Thus $u^* = m = \mathbb{E}[A \mid o]$.

DL §3.8; course derivation · Goodfellow et al. (2016)

EXAMPLE

Multimodal actions

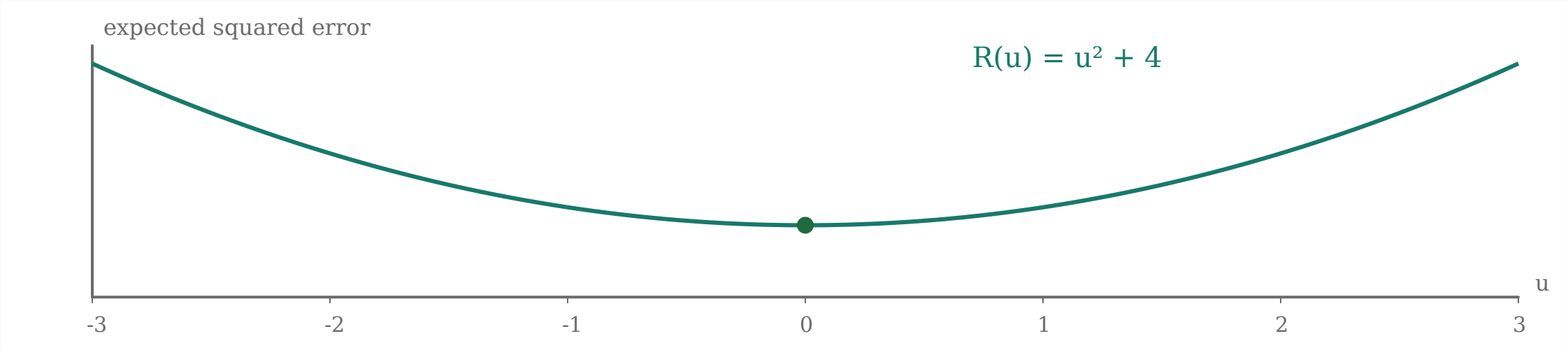
Worked example: the average misses both modes

EXAMPLE

$$R(u) = \tfrac{1}{2}(u+2)^2 + \tfrac{1}{2}(u-2)^2 = u^2 + 4$$

$R(u)$ expected squared error for A=−2 or +2, each with probability 1/2

u chosen scalar action



Diffusion Policy §4.1; course example · Chi et al. (2023/2024)

EXERCISE

Multimodal actions

Transfer: one route is more common

Find u, the prediction minimizing expected squared error. Compare it with the most probable action.

$$P(A = -2 \mid o) = \tfrac{3}{4}, \quad P(A = 2 \mid o) = \tfrac{1}{4}$$

- A random action at fixed observation o
- u deterministic prediction to optimize

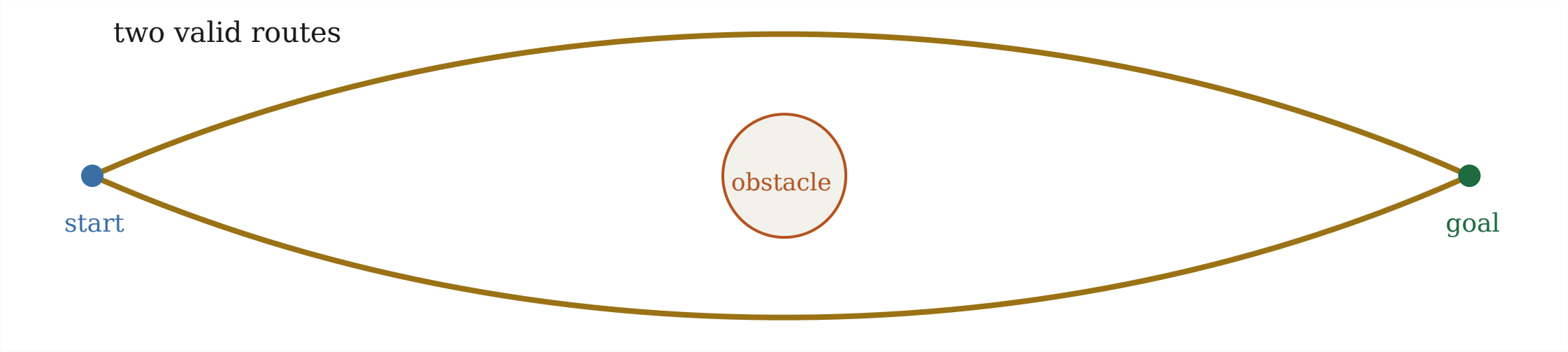
DL §3.8; course exercise · Goodfellow et al. (2016)

EXERCISE

Multimodal actions

A distribution and its mean answer different questions

Fixed observation; equally likely actions -2 and 2.
Compare: conditional mean vs a random draw.



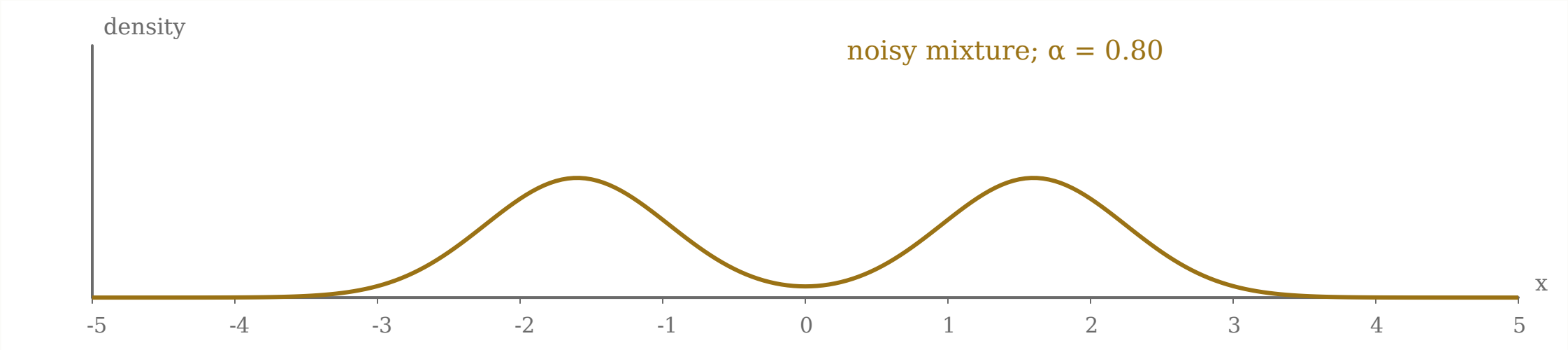
Diffusion Policy §4.1; course exercise · Chi et al. (2023/2024)

MOTIVATION

Forward diffusion

Noise creates a supervised learning task

Denoising estimates clean-data information from an artificially corrupted input.



DDPM §2, Algorithms 1–2 · Ho, Jain & Abbeel (2020)

MODEL

Forward diffusion

Two clocks and three kinds of vectors

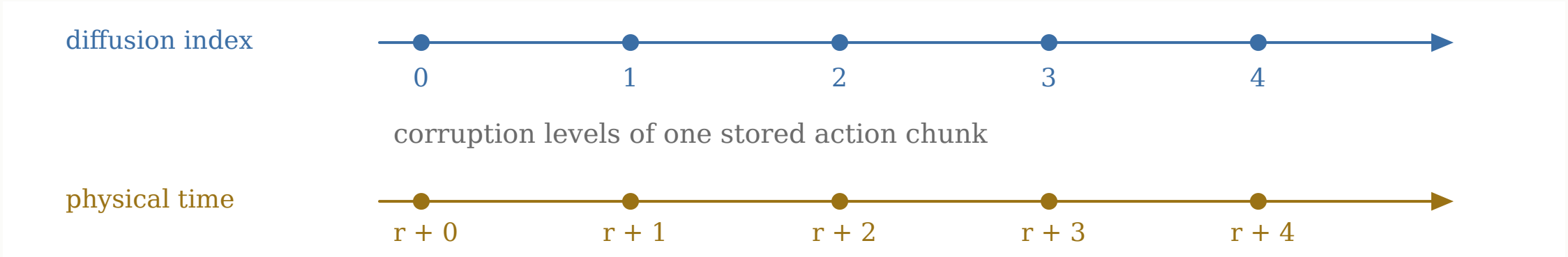
The diffusion index labels corruption levels of one stored vector; physical time labels robot execution.

$$x_0, x_k, \epsilon \in \mathbb{R}^d, \quad k \in \{1, \dots, K\}$$

$x_0, x_k, \epsilon \in \mathbb{R}^d$ clean vector, noisy vector, standard Gaussian noise; d components each

k, K diffusion level and final level

r physical robot time



DDPM §2; Diffusion Policy §2.3 · Ho, Jain & Abbeel (2020) · Chi et al. (2023/2024)

DEFINITION

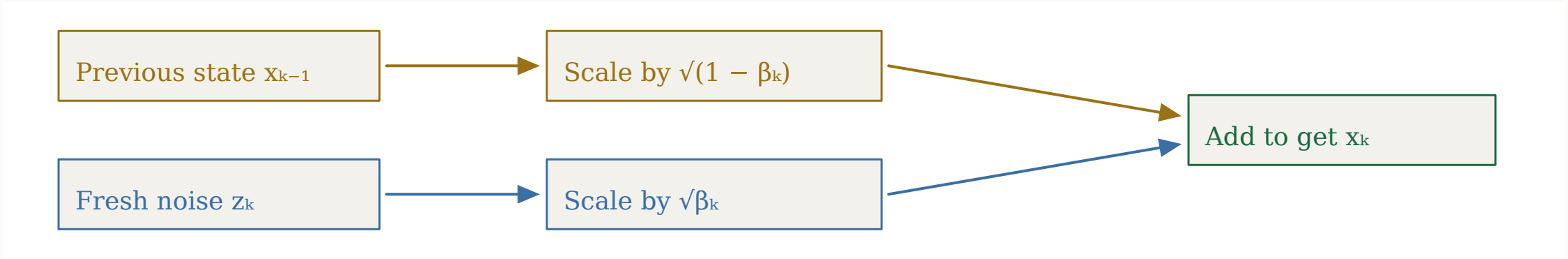
Forward diffusion

One step shrinks signal and adds fresh noise

Forward diffusion corrupts data one step at a time. Each step depends on the previous state and fresh noise.

$$x_k = \sqrt{1-\beta_k}\, x_{k-1} + \sqrt{\beta_k}\, z_k$$

x_{k-1}, x_k	previous and next vectors in $\mathbb{R}^d$
$0 < \beta_k < 1$	chosen step variance
$z_k \sim \mathcal{N}(0, I_d)$	fresh Gaussian noise, independent of earlier states; I_d is identity



DDPM §2, Eq.2 · Ho, Jain & Abbeel (2020)

EXERCISE

Forward diffusion

Check the conditional mean and covariance

Find the conditional mean and covariance of x_k.

$$x_k = \sqrt{1-\beta_k}\, x_{k-1} + \sqrt{\beta_k}\, z_k$$

$x_{k-1}, x_k \in \mathbb{R}^d$ fixed previous state and random next state

$0 < \beta_k < 1$ step variance

$z_k \sim \mathcal{N}(0, I_d)$ fresh independent noise; I_d is identity

DDPM Eq.2; DL §3.8 · Ho, Jain & Abbeel (2020) · Goodfellow et al. (2016)

EXERCISE

Forward diffusion

Transfer: variance is not the noise multiplier

Compute the realized x_k; show both coefficients.

$$x_k = \sqrt{1-\beta_k}\, x_{k-1} + \sqrt{\beta_k}\, z_k$$
$$\beta_k = 0.36, \quad x_{k-1} = 2, \quad z_k = -1$$

β_k variance

x_{k-1}, z_k given scalar values

DDPM Eq.2; course exercise · Ho, Jain & Abbeel (2020)

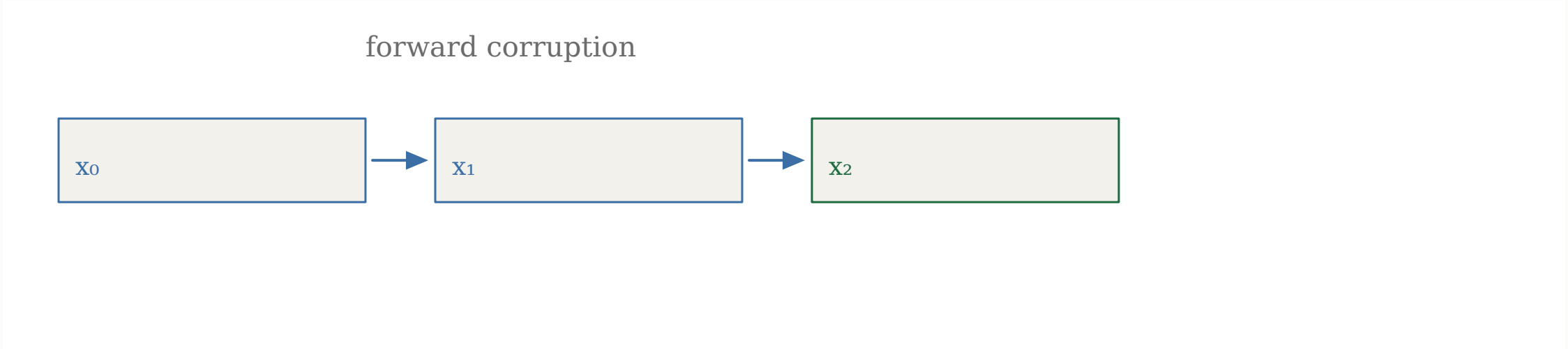
DERIVATION

Forward diffusion

Two steps expose the cumulative signal

$$x_2 = \sqrt{a_2 a_1}\, x_0 + \sqrt{a_2 \beta_1}\, z_1 + \sqrt{\beta_2}\, z_2$$

- x_0, x_2 clean and twice-corrupted vectors
- $a_j = 1 - \beta_j$ one-step retention; β_j is step variance
- z_1, z_2 independent standard Gaussian vectors



DDPM §2, Eq.4; course expansion · Ho, Jain & Abbeel (2020)

Forward diffusion

Two steps expose the cumulative signal

Substitute the first transition into the second

$a_j = 1 - \beta_j$. Independent z_1, z_2 are standard Gaussian vectors.

1. $x_1 = \sqrt{a_1}x_0 + \sqrt{\beta_1}z_1.$
2. $x_2 = \sqrt{a_2}x_1 + \sqrt{\beta_2}z_2.$
3. $x_2 = \sqrt{a_2}(\sqrt{a_1}x_0 + \sqrt{\beta_1}z_1) + \sqrt{\beta_2}z_2.$
4. $x_2 = \sqrt{a_2 a_1}x_0 + \sqrt{a_2\beta_1}z_1 + \sqrt{\beta_2}z_2.$

DDPM §2, Eq.4; course expansion · Ho, Jain & Abbeel (2020)

DERIVATION

Forward diffusion

Independent Gaussian variances combine

For fixed x_0, the random part of x_2 is $\sqrt{a_2\beta_1}\,z_1 + \sqrt{\beta_2}\,z_2$.

$$a_2\beta_1 + \beta_2 = (1-\beta_2)\beta_1 + \beta_2 = 1 - a_1a_2$$

$a_j = 1 - \beta_j$ retention; β_j is the added variance at step j

z_1, z_2 independent Gaussian vectors with identity covariance

DDPM Eq.4; course derivation · Ho, Jain & Abbeel (2020)

Forward diffusion

Independent Gaussian variances combine

Combine independent centered noise terms

The noise coefficients are $\sqrt{a_2\beta_1}$ and $\sqrt{\beta_2}$.

1. $\mathrm{Cov}(\sqrt{a_2\beta_1}z_1 + \sqrt{\beta_2}z_2) = (a_2\beta_1 + \beta_2)I.$
2. $a_2\beta_1 + \beta_2 = (1-\beta_2)\beta_1 + \beta_2 = 1 - a_1a_2.$
3. The sum is Gaussian. Replace it in distribution by a standard Gaussian scaled by sqrt(1−a1 a2).

DDPM Eq.4; course derivation · Ho, Jain & Abbeel (2020)

RESULT

Forward diffusion

Any noise level can be sampled directly

$$x_k = \alpha_k x_0 + \sigma_k \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

x_0, x_k	fixed clean, random noisy vectors
$\alpha_k = \sqrt{\bar{a}_k},\ \sigma_k = \sqrt{1 - \bar{a}_k}$	signal and noise scales
$\bar{a}_k = \prod_{j=1}^{k} a_j,\ a_j = 1 - \beta_j$	retentions; β_j step variance
ϵ	independent of x_0; I is identity

DDPM Eq.4; notation bridge · Ho, Jain & Abbeel (2020)

Forward diffusion

Any noise level can be sampled directly

Build the conditional law one level at a time

$a_j = 1 - \beta_j$; z_j are fresh standard Gaussian vectors; $\bar{a}_0 = 1$ and $v_0 = 0$.

1. Clean coefficient at level k: $\prod_{j=1}^{k} \sqrt{a_j} = \sqrt{\bar{a}_k}$.
2. Noise covariance is $v_k I$: $v_k = a_k v_{k-1} + \beta_k$.
3. If $v_{k-1} = 1 - \bar{a}_{k-1}$, then $v_k = 1 - a_k \bar{a}_{k-1} = 1 - \bar{a}_k$.
4. The noise remains Gaussian, so one draw scaled by $\sqrt{v_k}$ gives the level-k law.

DDPM Eq.4; notation bridge · Ho, Jain & Abbeel (2020)

INTERPRETATION

Forward diffusion

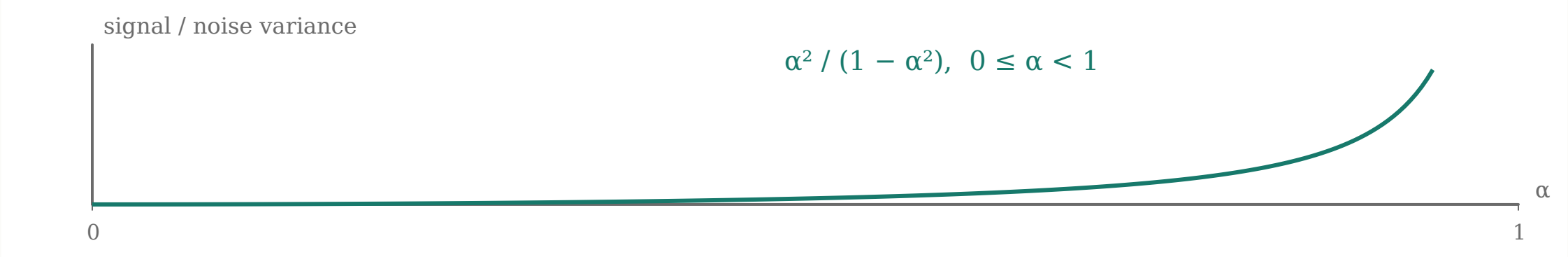
A schedule controls signal-to-noise ratio

For a unit-variance clean component, SNR is retained signal variance divided by added noise variance.

$$\mathrm{SNR}_k = \frac{\alpha_k^2}{\sigma_k^2} = \frac{\bar{a}_k}{1 - \bar{a}_k}$$

$\alpha_k, \sigma_k > 0$ signal coefficient and noise standard deviation

$\bar{a}_k = \alpha_k^2$ cumulative retention at diffusion level k



DDPM §4; Diffusion Policy §3.3 · Ho, Jain & Abbeel (2020) · Chi et al. (2023/2024)

EXAMPLE

Forward diffusion

Worked example: one noisy training pair

EXAMPLE

$$x_0 = 2,\ \alpha_k = 0.8,\ \sigma_k = 0.6,\ \epsilon = -0.5$$

$$x_k = \alpha_k x_0 + \sigma_k \epsilon = 1.3$$

$$\text{input} = (1.3, k), \quad \text{target} = -0.5$$

- x_0, x_k clean scalar and noisy scalar at level k
- ϵ known sampled noise and target
- α_k, σ_k fixed signal coefficient and noise standard deviation

DDPM Algorithm 1; course example · Ho, Jain & Abbeel (2020)

EXERCISE

Forward diffusion

Transfer: a two-component action

Compute x, then recover the sampled noise.

$$x_0 = \begin{bmatrix} 2 \\ -1 \end{bmatrix},\ \epsilon = \begin{bmatrix} -0.5 \\ 1 \end{bmatrix},\ \alpha = 0.8,\ \sigma = 0.6$$

x_0, ϵ two-coordinate clean and noise vectors

α, σ signal coefficient and noise standard deviation

$x = \alpha x_0 + \sigma \epsilon$ noisy vector to compute

DDPM Eq.4; course exercise · Ho, Jain & Abbeel (2020)

EXERCISE

Forward diffusion

Noisy covariance includes clean data spread

Find the mean and covariance.

$$X_k = \alpha_k X_0 + \sigma_k \epsilon, \quad X_0 \perp \epsilon$$

X_0, X_k	random clean and noisy vectors
μ_0, Σ_0	finite clean mean and covariance
α_k, σ_k	signal and noise scales
$\epsilon \sim \mathcal{N}(0, I_d)$	independent noise; I_d identity

DDPM Eq.4; DL §3.8 · Ho, Jain & Abbeel (2020) · Goodfellow et al. (2016)

ASSUMPTION

Forward diffusion

The terminal Gaussian is a schedule approximation

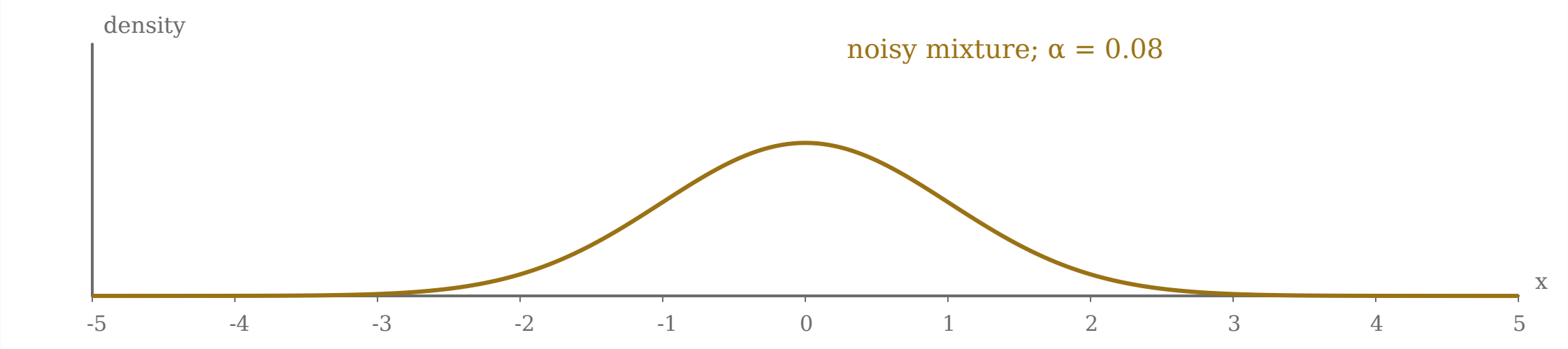
The generation prior is the simple distribution used to initialize the reverse sampler.

$$\alpha_K \approx 0, \quad \sigma_K \approx 1 \quad \Longrightarrow \quad X_K \approx \mathcal{N}(0, I)$$

K, X_K last diffusion level and its marginal noisy vector

α_K, σ_K final signal coefficient and noise standard deviation

I identity covariance



DDPM §4, terminal prior · Ho, Jain & Abbeel (2020)

EXERCISE

Forward diffusion

Does this terminal schedule match the prior?

Can the final noisy law be standard Gaussian?

$$\mu_K = \alpha_K \mu_0, \quad v_K = \alpha_K^2 v_0 + \sigma_K^2$$
$$\alpha_K = 0.8,\ \sigma_K = 0.6,\ \mu_0 = 5,\ v_0 = 1$$

μ_i, v_i scalar mean and variance at level i; 0 is clean

α_K, σ_K final signal and noise scales

DDPM §4; course exercise · Ho, Jain & Abbeel (2020)

DEFINITION

The score

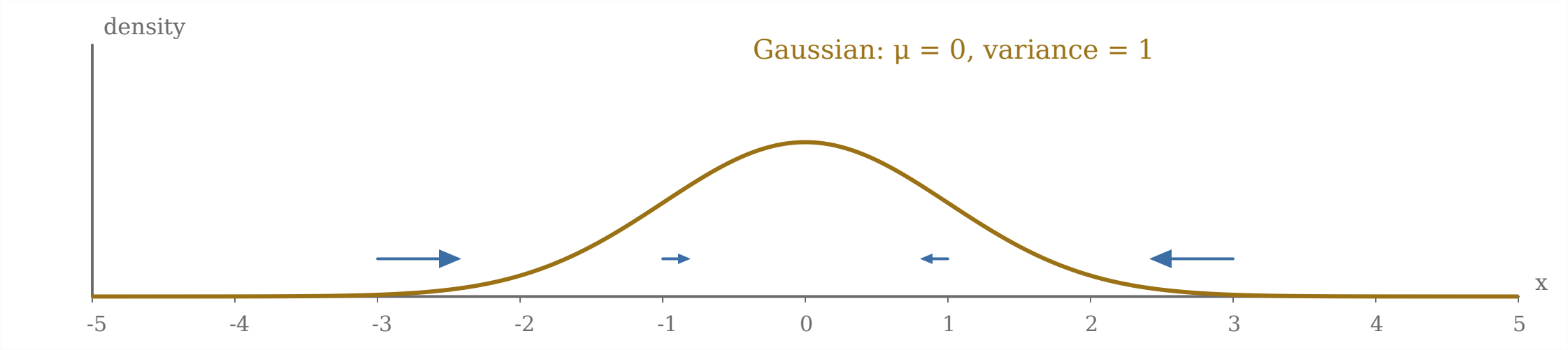
A density supplies a local direction

The score is the gradient of log density with respect to the data vector.

$$s_k(x) = \nabla_x \log p_k(x)$$

$p_k(x) > 0$ differentiable density at fixed noise level k

$s_k(x) \in \mathbb{R}^d$ input gradient; density parameters fixed



Hyvärinen (2005) §2, Eq.1; Vincent (2011) §3.1 · Vincent (2011) · Hyvärinen (2005)

MODEL

The score

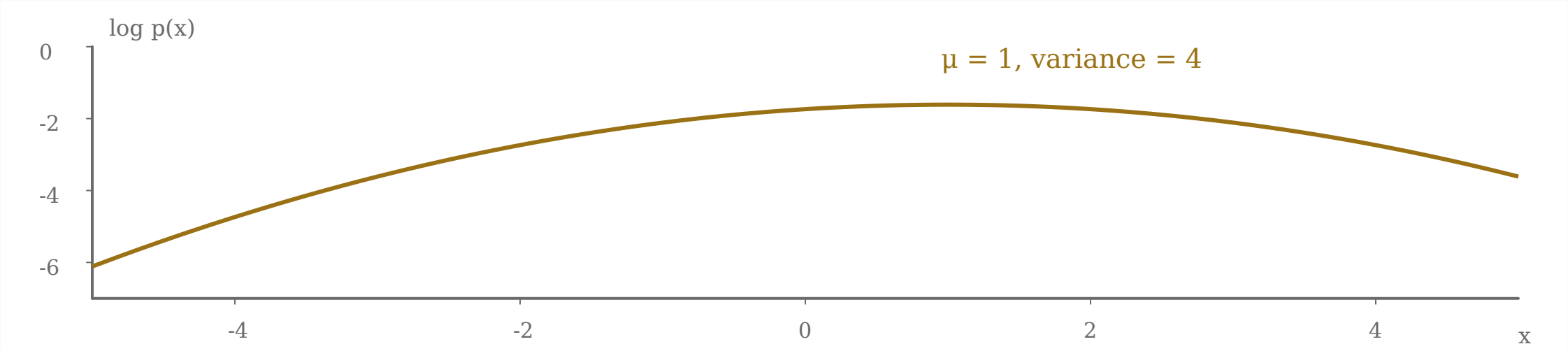
Start with a Gaussian log density

$$\log p(x) = C - \frac{(x-\mu)^2}{2v}$$

$x, p(x)$ scalar input and its Gaussian density

$\mu \in \mathbb{R},\ v > 0$ fixed mean and variance

$C = -\frac{1}{2}\log(2\pi v)$ constant in x



DL §3.9.3; course derivation · Goodfellow et al. (2016)

DERIVATION

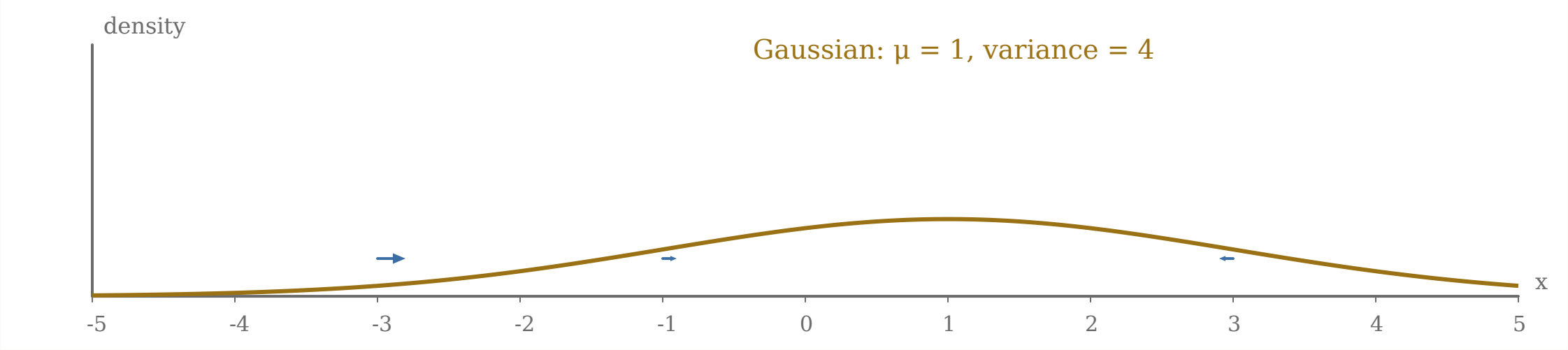
The score

Differentiate the quadratic one step at a time

Start with $\log p(x) = C - (x-\mu)^2/(2v)$, with C independent of x.

$$s(x) = -\frac{1}{2v}\,2(x-\mu) = -\frac{x-\mu}{v}$$

- μ, v fixed mean and positive variance
- $s(x)$ scalar derivative of log p with respect to x



DL §3.9.3; course derivation · Goodfellow et al. (2016)

The score

Differentiate the quadratic one step at a time

Differentiate the scalar Gaussian log density

μ and variance $v > 0$ are fixed; C is constant in x.

1. $\log p(x) = C - (x-\mu)^2/(2v)$.
2. $\partial_x C = 0$, and $\partial_x (x-\mu)^2 = 2(x-\mu)$.
3. $s(x) = -[2(x-\mu)]/(2v) = -(x-\mu)/v$.

DL §3.9.3; course derivation · Goodfellow et al. (2016)

RESULT

The score

Covariance shapes the vector direction

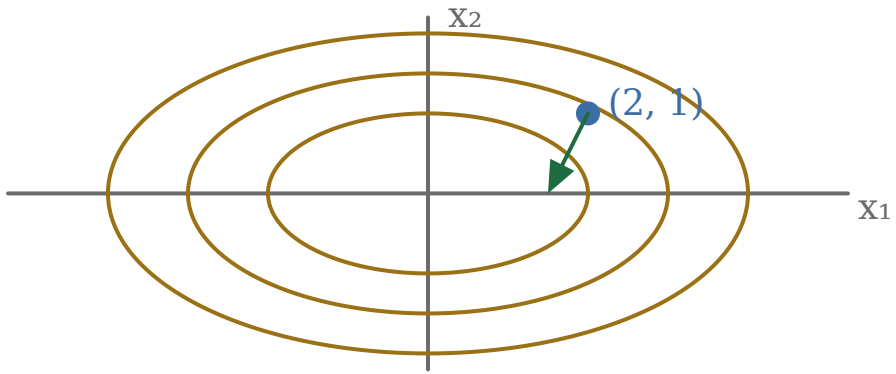
The precision matrix is the inverse covariance; it weights displacement in the Gaussian score.

$$s(x) = -\Sigma^{-1}(x - \mu)$$

$x, \mu \in \mathbb{R}^d$ state and mean

$\Sigma \succ 0$ fixed positive-definite covariance

$\Sigma = \mathrm{diag}(4, 1)$



DL §3.9.3; Lecture 5 quadratic differential · Goodfellow et al. (2016)

The score

Covariance shapes the vector direction

Differentiate a symmetric quadratic form

$r = x - \mu$ is a column vector; positive-definite covariance Σ is fixed.

1. $\log p(x) = C - \frac{1}{2} r^T \Sigma^{-1} r.$
2. $d(r^T \Sigma^{-1} r) = dr^T \Sigma^{-1} r + r^T \Sigma^{-1} dr = 2 r^T \Sigma^{-1} dx.$
3. $d \log p(x) = (-\Sigma^{-1} r)^T dx$, so $s(x) = -\Sigma^{-1}(x - \mu).$

DL §3.9.3; Lecture 5 quadratic differential · Goodfellow et al. (2016)

EXERCISE

The score

Compute an anisotropic score

The Gaussian has different variances along the two axes. Find the score and compare its direction with −x.

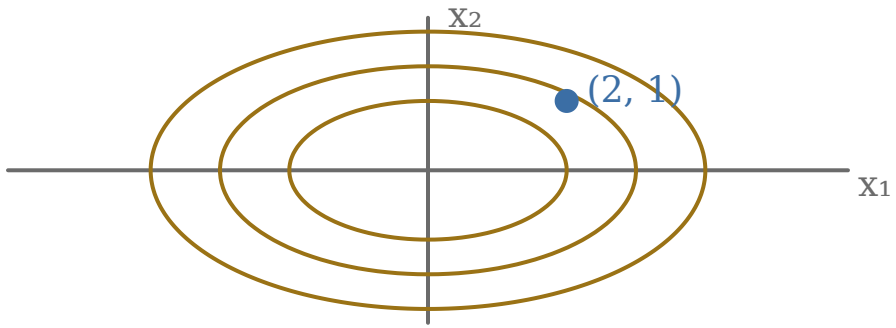
$$\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \; \Sigma = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}, \; x = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

x, μ evaluation point and mean

Σ covariance matrix

$s(x) = -\Sigma^{-1}(x - \mu)$ score to compute

$\Sigma = \mathrm{diag}(4, 1)$



DL §3.9.3; course exercise · Goodfellow et al. (2016)

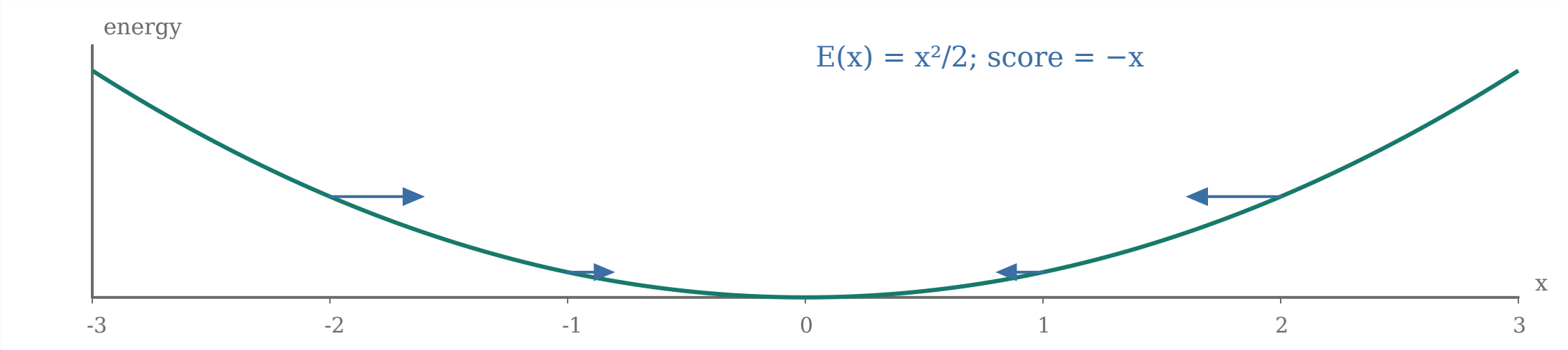
DERIVATION

The score

An unknown normalizing constant disappears

$$p(x) = \frac{e^{-E(x)}}{Z} \quad \Longrightarrow \quad \nabla_x \log p(x) = -\nabla_x E(x)$$

- $x, p(x)$ input vector and its density
- $E(x)$ differentiable energy
- Z finite positive normalization independent of x



Hyvärinen (2005) §2, Eq.1; course energy form · Hyvärinen (2005)

DERIVATION

The score

A mixture score combines component scores

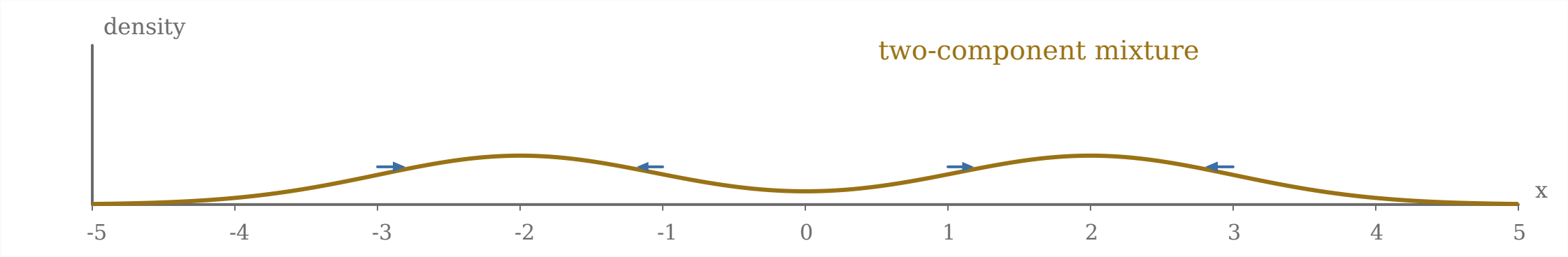
A mixture averages densities. Responsibility is a component’s posterior weight at x.

$$s(x) = \sum_j r_j(x) s_j(x), \quad r_j(x) = \frac{w_j p_j(x)}{\sum_i w_i p_i(x)}$$

$w_j \geq 0,\ \sum_j w_j = 1$ fixed mixture weights

p_j, s_j positive component density and its score

$r_j(x)$ posterior component probability at x



DL §3.9.6; course derivation · Goodfellow et al. (2016)

The score

A mixture score combines component scores

Differentiate the sum before taking responsibilities

Fixed weights $w_j \geq 0$ sum to one; component densities are differentiable.

1. $p(x) = \sum_j w_j p_j(x)$, so $\nabla p(x) = \sum_j w_j \nabla p_j(x)$.
2. $s(x) = \nabla p(x)/p(x) = \sum_j [w_j p_j(x)/p(x)] \, \nabla \log p_j(x)$.
3. Set $r_j(x) = w_j p_j(x)/p(x)$ to obtain $s(x) = \sum_j r_j(x) s_j(x)$.

DL §3.9.6; course derivation · Goodfellow et al. (2016)

EXERCISE

The score

What does a stationary point tell us?

Find s(0). Is x=0 a mode? Explain.

$$p(x) = \tfrac{1}{2}\mathcal{N}(x; -2, 1) + \tfrac{1}{2}\mathcal{N}(x; 2, 1)$$

x	scalar input
$\mu_1 = -2,\ \mu_2 = 2$	component means; variances are 1
$s_j(x) = -(x - \mu_j)$	component score
$s(x) = \sum_j r_j(x) s_j(x)$	mixture score; r_j is posterior component weight

DL §3.9.6; course exercise · Goodfellow et al. (2016)

EXERCISE

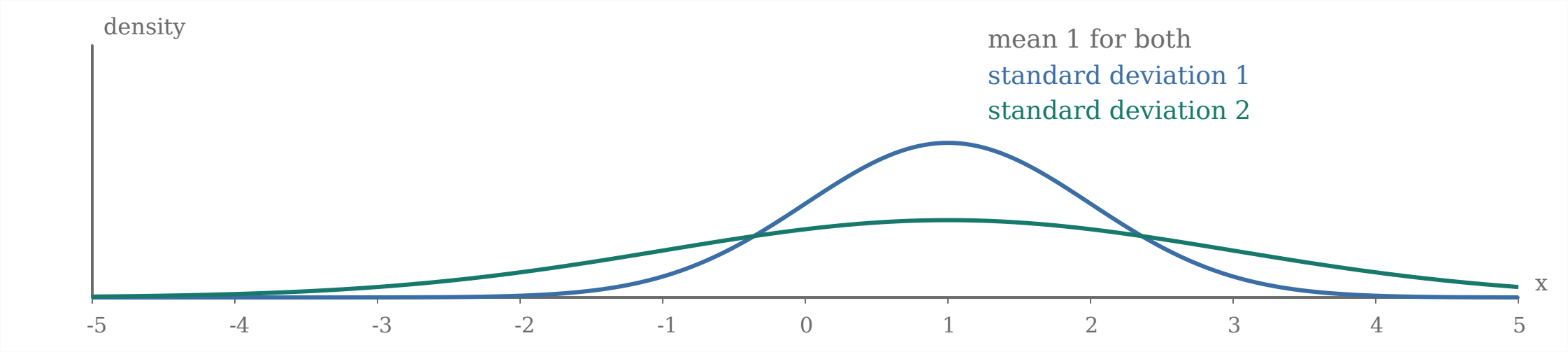
The score

Transfer: doubling standard deviation

Compare standard deviations 1 and 2. Compute both scores.

$$\mu = 1, \quad x = 3, \quad s(x) = -(x - \mu)/v$$

- μ, x mean and evaluation point
- $v, \sqrt{v}$ variance and standard deviation
- $s(x)$ score to compute



DL §3.9.3; course exercise · Goodfellow et al. (2016)

MODEL

Learning to denoise

Known corruption gives a target for every example

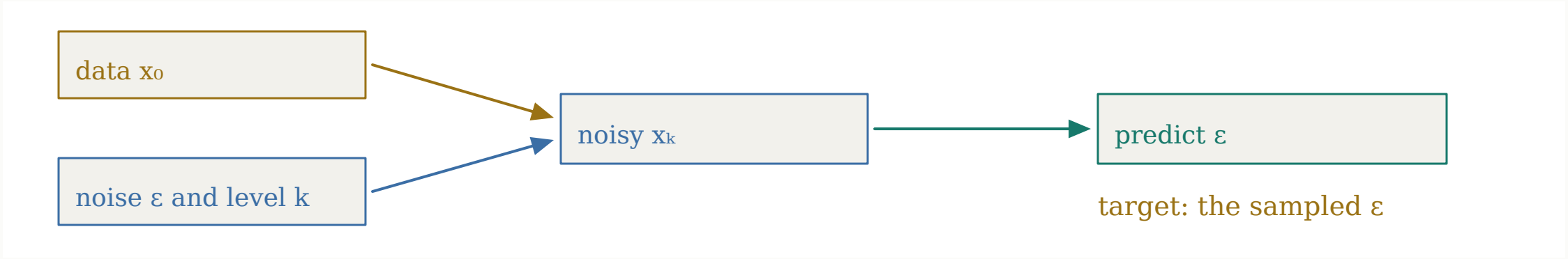
The noise drawn for corruption becomes the target.

$$(x_0, k, \epsilon) \longmapsto \big((x_k, k), \epsilon\big)$$

x_0, k clean demonstration and sampled level

$x_k = \alpha_k x_0 + \sigma_k \epsilon$ noisy vector; α_k, σ_k are signal and noise scales

ϵ sampled Gaussian noise and target



DDPM Algorithm 1 · Ho, Jain & Abbeel (2020)

DEFINITION

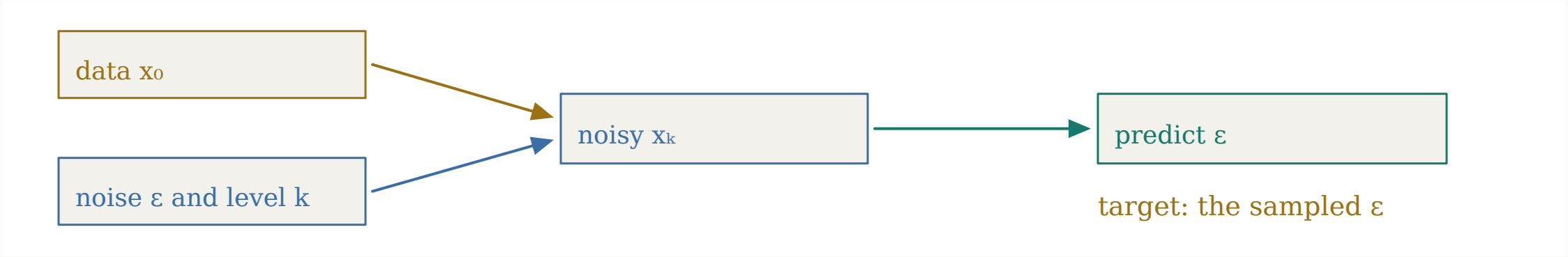
Learning to denoise

The simple noise-prediction objective

The loss averages squared prediction errors.

$$L(\theta) = \mathbb{E}\|\epsilon_\theta(\alpha_k x_0 + \sigma_k \epsilon, k) - \epsilon\|^2$$

- x_0, k clean data and sampled level
- α_k, σ_k fixed signal and noise scales
- $\theta, \epsilon_\theta, \epsilon$ weights, predicted noise, sampled noise
- $\mathbb{E}$ average over data, uniform k and independent Gaussian noise



DDPM §3.4, Eq.14 · Ho, Jain & Abbeel (2020)

DERIVATION

Learning to denoise

The optimal predictor is a conditional mean

Finite conditional noise second moment.

$$\mathbb{E}[\|\epsilon - g\|^2 \mid x_k, k] = \mathbb{E}[\|\epsilon - m\|^2 \mid x_k, k] + \|g - m\|^2$$

x_k, k, ϵ fixed noisy vector and level; random noise target

$m = \mathbb{E}[\epsilon \mid x_k, k]$ conditional mean

g candidate noise prediction

DDPM Eq.14; course least-squares derivation · Ho, Jain & Abbeel (2020)

Learning to denoise

The optimal predictor is a conditional mean

Expand the vector squared distance

Fix x_k, k. Let $m = \mathbb{E}[\epsilon \mid x_k, k]$; assume a finite conditional second moment.

1. $\epsilon - g = (\epsilon - m) + (m - g)$.
2. $\|\epsilon - g\|^2 = \|\epsilon - m\|^2 + 2(\epsilon - m)^T(m - g) + \|m - g\|^2$.

DDPM Eq.14; course least-squares derivation · Ho, Jain & Abbeel (2020)

Learning to denoise

The optimal predictor is a conditional mean

Average and minimize

m, g are fixed under this conditional expectation.

1. $\mathbb{E}[(\epsilon - m)^T (m - g) \mid x_k, k] = 0.$
2. $\mathbb{E}[\|\epsilon - g\|^2 \mid x_k, k] = \mathbb{E}[\|\epsilon - m\|^2 \mid x_k, k] + \|g - m\|^2.$
3. Only the last term depends on g, so $g^* = m.$

DDPM Eq.14; course least-squares derivation · Ho, Jain & Abbeel (2020)

DERIVATION

Learning to denoise

One clean example has an accessible conditional score

Fixed x_0: mean $\alpha_k x_0$, covariance $\sigma_k^2 I$; I is identity.

$$\nabla_x \log q_k(x \mid x_0) = -\frac{x - \alpha_k x_0}{\sigma_k^2} = -\frac{\epsilon}{\sigma_k}$$

$q_k(x \mid x_0)$	corruption density
$x = \alpha_k x_0 + \sigma_k \epsilon$	noisy vector; ε is standard Gaussian noise
x_0, k	fixed clean vector and level
$\alpha_k, \sigma_k > 0$	signal and noise scales

Vincent (2011) §4.2, Eq.4.4; Gaussian score derivation · Vincent (2011)

DERIVATION

Learning to denoise

Average conditional scores to obtain the score

Clean density: integrate. Discrete clean data: sum.

$$\nabla_x \log p_k(x) = \int p_k(x_0 \mid x) \nabla_x \log q_k(x \mid x_0)\, dx_0$$

$q_k(x \mid x_0)$ corruption density

$p_k(x) > 0$ noisy marginal; derivative passes under integral

$p_k(x_0 \mid x)$ clean posterior at level k

Vincent (2011) Appendix A, Eqs.A.1–A.2; course Bayes derivation · Vincent (2011)

Learning to denoise

Average conditional scores to obtain the score

Differentiate the noisy marginal density

$q_k(x \mid x_0)$ is the Gaussian corruption density; p_0 is the clean density; fix k.

1. $p_k(x) = \int p_0(x_0) q_k(x \mid x_0)\, dx_0$.
2. $\nabla_x p_k(x) = \int p_0(x_0) \nabla_x q_k(x \mid x_0)\, dx_0$.
3. The second equality assumes the derivative can pass under the integral.

Vincent (2011) Appendix A, Eqs.A.1–A.2; course Bayes derivation · Vincent (2011)

Learning to denoise

Average conditional scores to obtain the score

Divide by the marginal and apply Bayes rule

Assume $p_k(x) > 0$. Bayes rule gives $p_k(x_0 \mid x) = p_0(x_0) q_k(x \mid x_0) / p_k(x)$.

1. $\nabla_x \log p_k(x) = \int \frac{p_0(x_0) q_k(x|x_0)}{p_k(x)} \frac{\nabla_x q_k(x|x_0)}{q_k(x|x_0)} \, dx_0.$
2. $s_k(x) = \mathbb{E}[\nabla_x \log q_k(x \mid X_0) \mid X_k = x].$
3. For discrete clean support, replace the integrals by sums.

Vincent (2011) Appendix A, Eqs.A.1–A.2; course Bayes derivation · Vincent (2011)

EXERCISE

Learning to denoise

Noise prediction gives a score estimate

Explain the minus sign and the factor 1/σ_k.

$$s_k(x) = -\frac{\mathbb{E}[\epsilon \mid x_k = x, k]}{\sigma_k} \approx -\frac{\epsilon_\theta(x, k)}{\sigma_k}$$

- $s_k(x)$ — true score at noisy input x and fixed level k
- $\epsilon, \epsilon_\theta(x, k)$ — sampled noise and network prediction
- $\sigma_k > 0$ — noise standard deviation

Vincent (2011) §4.2, Eqs.4.3–4.5; DDPM §3.2 · Ho, Jain & Abbeel (2020) · Vincent (2011)

EXAMPLE

Learning to denoise

Worked example: noise prediction becomes a score

EXAMPLE

$$\sigma_k = 0.6, \quad \epsilon_\theta = (-0.3, 0.6)^T$$

$$s_\theta = -\epsilon_\theta / \sigma_k = (0.5, -1)^T$$

$\epsilon_\theta, \sigma_k$ predicted noise and noise standard deviation

$s_\theta = -\epsilon_\theta / \sigma_k$ score estimate at this input

DDPM §3.2; course example · Ho, Jain & Abbeel (2020)

EXERCISE

Learning to denoise

What can a zero noise prediction mean?

Find the noise for each clean hypothesis, then its mean.

$$X_0 \in \{-1, 1\}, \quad \alpha = \sigma = 1/\sqrt{2}, \quad X_k = 0$$

- X_0 — clean value, -1 or $+1$ with equal probability
- α, σ — signal and noise scales
- $X_k = \alpha X_0 + \sigma \epsilon$ — noisy value; independent standard Gaussian ε; condition via density

DDPM Eq.14; course exercise · Ho, Jain & Abbeel (2020)

RESULT

Learning to denoise

A denoising identity gives the posterior mean

Scaled Tweedie formula: recover the mean clean vector from the noisy score.

$$\mathbb{E}[X_0 \mid X_k = x] = \frac{x + \sigma_k^2 s_k(x)}{\alpha_k}$$

X_0, X_k, x	clean random vector, noisy random vector, observation
$\alpha_k, \sigma_k > 0$	positive signal and noise scales
$s_k(x)$	true noisy score

Efron (2011), Eq.1.4; scaled Tweedie identity · Efron (2011)

Learning to denoise

A denoising identity gives the posterior mean

Invert the known corruption algebraically

$X_k = \alpha_k X_0 + \sigma_k \epsilon$, with $\alpha_k, \sigma_k > 0$; condition on $X_k = x$.

1. $X_0 = (x - \sigma\epsilon)/\alpha.$
2. $\mathbb{E}[X_0 \mid X_k = x] = (x - \sigma\mathbb{E}[\epsilon \mid X_k = x])/\alpha.$
3. The observed x is fixed inside this conditional expectation.

Efron (2011), Eq.1.4; scaled Tweedie identity · Efron (2011)

Learning to denoise

A denoising identity gives the posterior mean

Substitute the noise-score relation

The exact marginal score obeys $\mathbb{E}[\epsilon \mid X_k = x] = -\sigma_k s_k(x)$.

1. $\mathbb{E}[X_0 \mid x] = (x - \sigma[-\sigma s_k(x)])/\alpha.$
2. $\mathbb{E}[X_0 \mid x] = (x + \sigma^2 s_k(x))/\alpha.$
3. This is a posterior mean, not a random sample or a guaranteed mode.

Efron (2011), Eq.1.4; scaled Tweedie identity · Efron (2011)

EXAMPLE

Learning to denoise

Worked example: two equivalent denoising routes

EXAMPLE

$$\hat{x}_0 = \frac{x - \sigma\hat{\epsilon}}{\alpha} = \frac{x + \sigma^2 s_\theta}{\alpha} = 2$$

$x = 1.3,\ \alpha = 0.8,\ \sigma = 0.6$ current scalar input and coefficients

$\hat{\epsilon} = -0.5$ predicted noise

$s_\theta = -\hat{\epsilon}/\sigma = 5/6$ equivalent score estimate

DDPM Eq.11; Efron Eq.1.4; course example · Ho, Jain & Abbeel (2020) · Efron (2011)

EXERCISE

Learning to denoise

How does predictor error reach the clean estimate?

Find the clean-estimate error. Compare $\alpha = 0.8, 0.1$.

$$\hat{x}_0 = \frac{x - \sigma\hat{\epsilon}}{\alpha}, \quad \hat{\epsilon} = \epsilon^* + \delta$$

- x fixed noisy input
- $\epsilon^*, \hat{\epsilon}, \delta$ optimal mean, prediction, error
- $\sigma = \sqrt{1 - \alpha^2}$ noise scale; α > 0 is signal scale

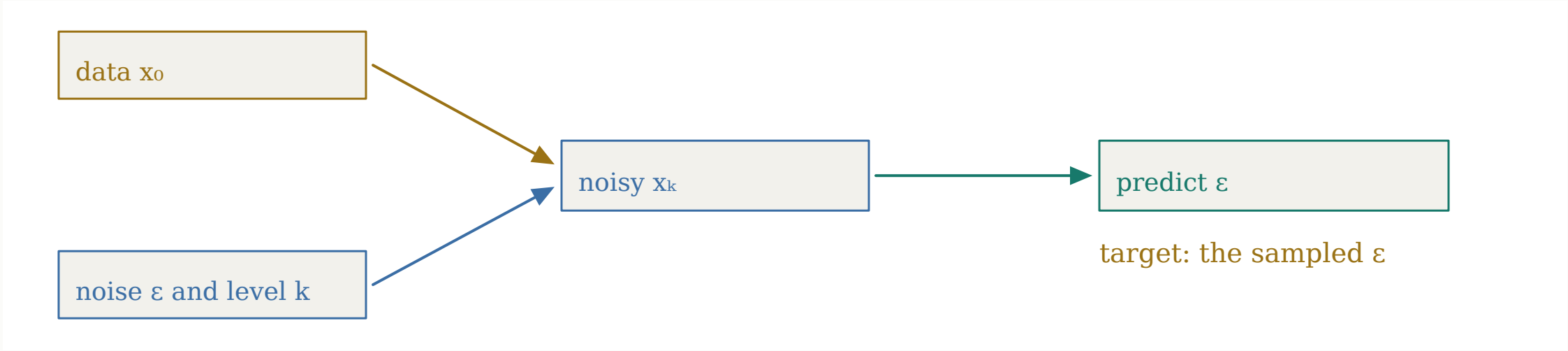
DDPM Eq.11; course exercise · Ho, Jain & Abbeel (2020)

ALGORITHM

Learning to denoise

Training updates weights, not the noisy input

Draw clean data x_0, level k and noise ε. Form the noisy input x_k. Compare the prediction with ε and update the denoiser weights θ.



DDPM Algorithm 1 · Ho, Jain & Abbeel (2020)

EXERCISE

Learning to denoise

Checkpoint: trace an input to a score

Trace one training example. Identify the sampled quantities, the network input, and the prediction target. How does the best prediction give a score?

DDPM Algorithm 1; Vincent (2011) §4.2, Eqs.4.3–4.5 · Ho, Jain & Abbeel (2020) · Vincent (2011)

MODEL

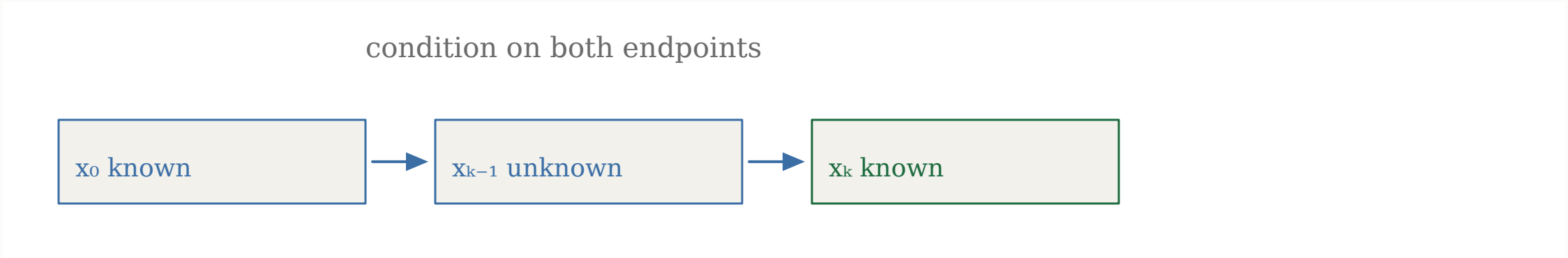
Reverse sampling

The reverse step needs a conditional distribution

To derive a reverse step, condition on both the current noisy state x_k and its clean source x_0.

$$q(x_{k-1} \mid x_k, x_0) \propto q(x_k \mid x_{k-1}) q(x_{k-1} \mid x_0)$$

- q forward-chain density or conditional density
- x_{k-1} unknown previous vector
- x_0, x_k fixed clean and noisy endpoints; k > 1
- $\propto$ proportional in the unknown state



DDPM §2, Eqs.6–7 · Ho, Jain & Abbeel (2020)

DERIVATION

Reverse sampling

The Gaussian product adds two quadratic penalties

$$-2\log q(y \mid x_k, x_0) = \frac{\|y - \alpha_{k-1}x_0\|^2}{\sigma_{k-1}^2} + \frac{\|x_k - \sqrt{a_k}y\|^2}{\beta_k} + C$$

- $y = x_{k-1}$ unknown previous vector; k > 1
- x_0, x_k fixed clean and noisy endpoints
- $\alpha_{k-1}, \sigma_{k-1}$ previous signal/noise scales
- $a_k = 1 - \beta_k$ step retention; β_k is step variance
- C constant in y

DDPM Appendix A, posterior derivation · Ho, Jain & Abbeel (2020)

DERIVATION

Reverse sampling

Complete the square

Fix x_0, x_k, with $k > 1$. Complete the square in the previous state $y = x_{k-1}$.

$$\Lambda \|y\|^2 - 2h^T y + C$$

ΛI, h posterior precision and linear coefficient vector

C constant in y

DDPM Eqs.6–7; Appendix A · Ho, Jain & Abbeel (2020)

Reverse sampling

Complete the square

Expand and collect terms in the unknown state

$y = x_{k-1}$; x_0, x_k are fixed and $\sigma_{k-1}^2 > 0$.

1. $\Lambda = 1/\sigma_{k-1}^2 + a_k/\beta_k.$
2. $h = \alpha_{k-1} x_0/\sigma_{k-1}^2 + \sqrt{a_k} x_k/\beta_k.$
3. $-2 \log q(y \mid x_k, x_0) = \Lambda \|y\|^2 - 2h^T y + C.$

DDPM Eqs.6–7; Appendix A · Ho, Jain & Abbeel (2020)

Reverse sampling

Complete the square

Complete the square and read the Gaussian

$\Lambda > 0$ is a scalar precision; h is a column vector.

1. $\Lambda\|y\|^2 - 2h^T y + C = \Lambda\|y - h/\Lambda\|^2 + C'$.
2. $\tilde{\mu} = h/\Lambda$, $\tilde{\beta} = 1/\Lambda$.
3. $q(y \mid x_k, x_0) = \mathcal{N}(y; \tilde{\mu}, \tilde{\beta} I)$.

DDPM Eqs.6–7; Appendix A · Ho, Jain & Abbeel (2020)

RESULT

Reverse sampling

Simplify the posterior coefficients

$$\tilde{\mu}_k = c_0 x_0 + c_k x_k, \quad \tilde{\beta}_k = \frac{\beta_k(1 - \bar{a}_{k-1})}{1 - \bar{a}_k}$$

- x_0, x_k fixed clean and noisy endpoints; k > 1
- a_k, β_k step retention and variance
- $\bar{a}_i$ cumulative retention through level i
- $\tilde{\mu}_k, \tilde{\beta}_k$ posterior mean and variance
- c_0, c_k clean and noisy coefficients

DDPM Eq.7 · Ho, Jain & Abbeel (2020)

Reverse sampling

Simplify the posterior coefficients

Simplify the precision denominator

$a_k = 1 - \beta_k$; $\bar{a}_k = a_k \bar{a}_{k-1}$. Λ is precision, h the linear coefficient vector, and α_{k-1} the previous signal amplitude.

1. $\Lambda = \frac{\beta_k + a_k(1-\bar{a}_{k-1})}{\beta_k(1-\bar{a}_{k-1})}$.
2. $\beta_k + a_k(1 - \bar{a}_{k-1}) = 1 - \bar{a}_k$.
3. $\tilde{\beta}_k = 1/\Lambda = \beta_k(1 - \bar{a}_{k-1})/(1 - \bar{a}_k)$.
4. $\tilde{\mu}_k = \tilde{\beta}_k h = c_0 x_0 + c_k x_k$, where $c_0 = \beta_k \alpha_{k-1}/(1 - \bar{a}_k)$ and $c_k = \sqrt{a_k}(1 - \bar{a}_{k-1})/(1 - \bar{a}_k)$.

DDPM Eq.7 · Ho, Jain & Abbeel (2020)

DERIVATION

Reverse sampling

Replace clean data by the learned noise estimate

Average over clean sources.

$$\mu_\theta(x_k, k) = \frac{x_k - \beta_k \hat{\epsilon}/\sigma_k}{\sqrt{a_k}}$$

$x_k, \hat{\epsilon}$ current state and predicted noise

$a_k = 1 - \beta_k$ step retention; β_k is step variance

σ_k cumulative noise scale

μ_θ reverse mean from denoiser weights θ

DDPM §3.2, Eq.11 · Ho, Jain & Abbeel (2020)

Reverse sampling

Replace clean data by the learned noise estimate

Insert the clean estimate into the known-clean mean

$\alpha_k = \sqrt{a_k}\alpha_{k-1}$; $\sigma_k^2 = 1 - \bar{a}_k$.

1. $\hat{x}_0 = (x_k - \sigma_k \hat{\epsilon})/\alpha_k$.
2. $\mu_\theta = \frac{\beta_k \alpha_{k-1}}{\sigma_k^2}\hat{x}_0 + \frac{\sqrt{a_k}(1-\bar{a}_{k-1})}{\sigma_k^2}x_k$.
3. The noise coefficient becomes $-\beta_k/(\sqrt{a_k}\sigma_k)$.

DDPM §3.2, Eq.11 · Ho, Jain & Abbeel (2020)

Reverse sampling

Replace clean data by the learned noise estimate

Simplify the remaining state coefficient

$\beta_k + a_k(1 - \bar{a}_{k-1}) = \sigma_k^2.$

1. Coefficient on x_k: $[\beta_k + a_k(1 - \bar{a}_{k-1})]/(\sqrt{a_k}\sigma_k^2) = 1/\sqrt{a_k}$.
2. $\mu_\theta = (x_k - \beta_k\hat{\epsilon}/\sigma_k)/\sqrt{a_k}$.
3. With optimal mean-noise prediction this gives the reverse conditional mean; a fixed-variance Gaussian still approximates its full law.

DDPM §3.2, Eq.11 · Ho, Jain & Abbeel (2020)

EXAMPLE

Reverse sampling

Worked example: calculate one reverse mean

EXAMPLE

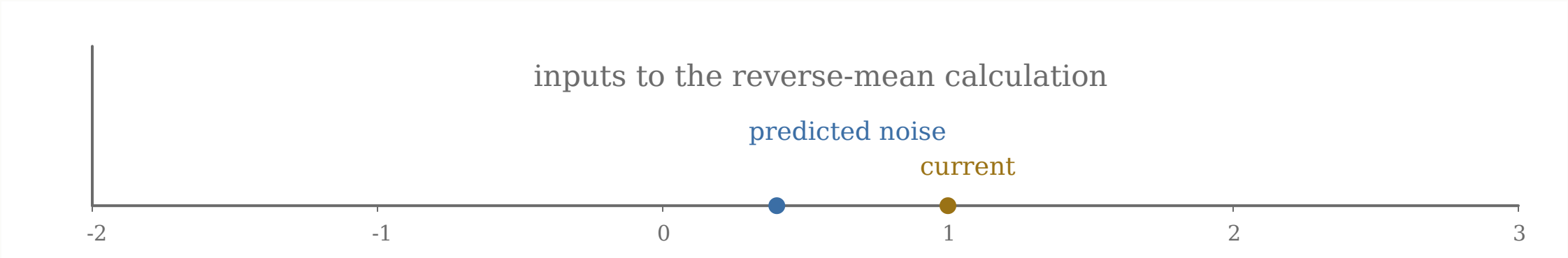
$$\mu_\theta = \frac{x_k - \beta_k \hat{\epsilon}/\sigma_k}{\sqrt{a_k}} = \frac{1 - 0.36(0.4)/0.8}{0.8} = 1.025$$

$\beta_k = 0.36,\ a_k = 0.64$ step variance and retention

$\sigma_k = 0.8$ cumulative noise standard deviation

$x_k = 1,\ \hat{\epsilon} = 0.4$ current state and predicted noise

μ_θ reverse mean



DDPM Eq.11; course example · Ho, Jain & Abbeel (2020)

EXERCISE

Reverse sampling

A reverse draw uses a standard deviation

Find the variance and one draw.

$$x_{k-1} = \mu_\theta + \sqrt{\tilde{\beta}_k}\, z, \quad \tilde{\beta}_k = \beta_k(1 - \bar{a}_{k-1})/(1 - \bar{a}_k)$$

$\mu_\theta = 1.025,\ z = -1$	mean and standard-normal draw
$\beta_k = 0.36$	step variance
$\bar{a}_k = 0.36,\ \bar{a}_{k-1} = 0.5625$	retentions; k > 1

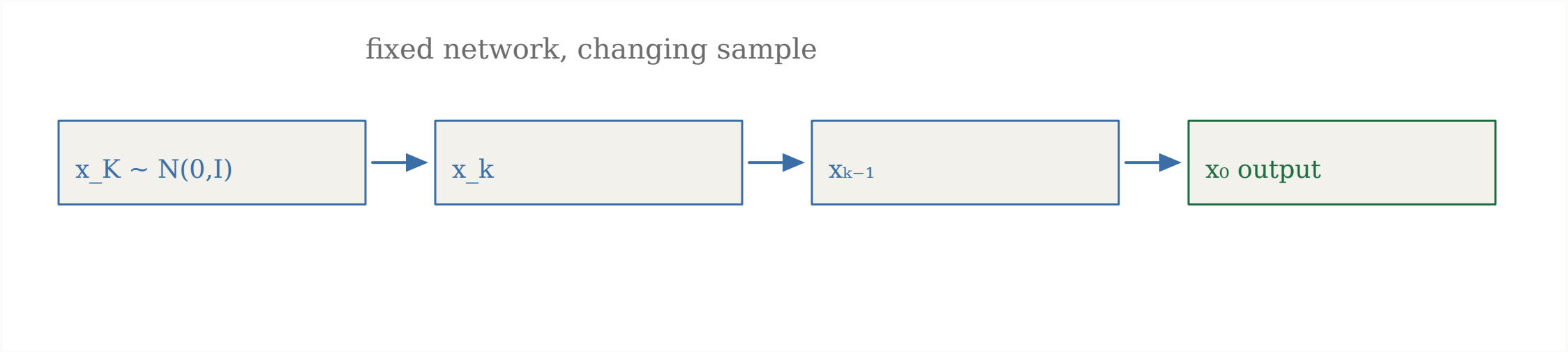
DDPM Algorithm 2; course exercise · Ho, Jain & Abbeel (2020)

ALGORITHM

Reverse sampling

DDPM generation runs from noise to data

Denoising diffusion probabilistic model (DDPM): start with Gaussian noise. Predict a reverse mean and add Gaussian noise at each step until the clean endpoint.



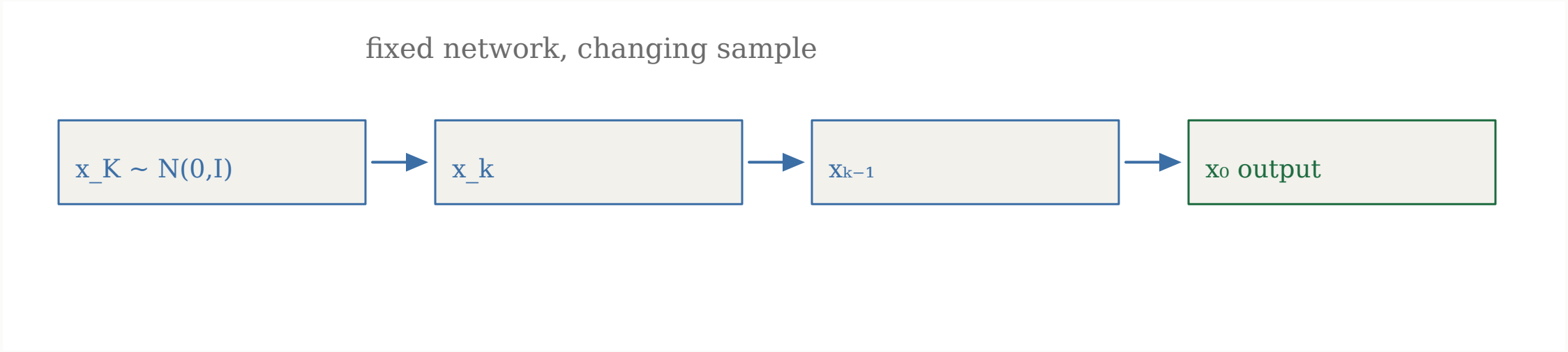
DDPM Algorithm 2; continuous-action adaptation · Ho, Jain & Abbeel (2020)

EXERCISE

Reverse sampling

Does a mean predictor force identical samples?

Same observation and trained weights.
Different initial Gaussian draws.
Must the final samples be identical?



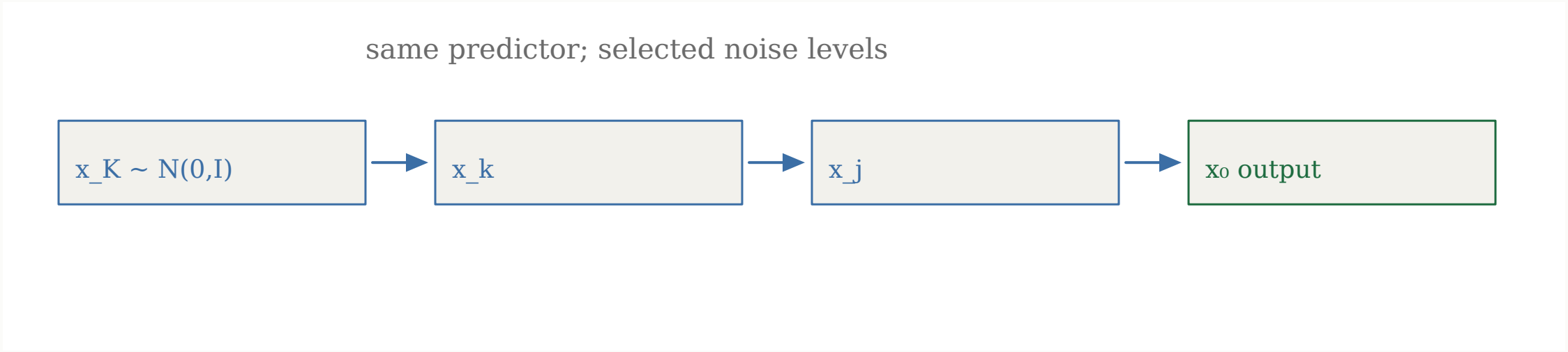
DDPM Algorithm 2; course exercise · Ho, Jain & Abbeel (2020)

MOTIVATION

Reverse sampling

DDIM changes the path, using the same predictor

A denoising diffusion implicit model (DDIM) can skip noise levels and denoise deterministically once the initial draw is fixed.



DDIM §§3–4 · Song, Meng & Ermon (2021)

RESULT

Reverse sampling

A deterministic DDIM step keeps the predicted direction

$$\hat{x}_0 = \frac{x_k - \sigma_k \hat{\epsilon}}{\alpha_k}, \qquad x_j = \alpha_j \hat{x}_0 + \sigma_j \hat{\epsilon}$$

$x_k, x_j, \hat{x}_0$	current state, destination state and clean estimate
$0 \le j < k$	select a less noisy level
$\hat{\epsilon} = \epsilon_\theta(x_k, k)$	predicted noise
$\alpha_i > 0,\ \sigma_i \ge 0$	signal and noise coefficients; α_i² + σ_i² = 1

DDIM §4.1, Eq.12 with eta=0 · Song, Meng & Ermon (2021)

EXAMPLE

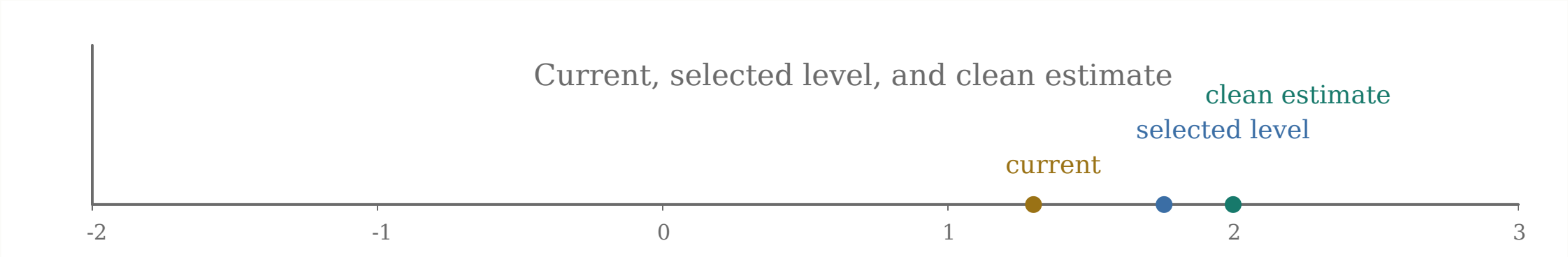
Reverse sampling

Worked example: skip to a cleaner level

EXAMPLE

$$\hat{x}_0 = \frac{1.3 - 0.6(-0.5)}{0.8} = 2, \qquad x_j = \sqrt{0.91}(2) + 0.3(-0.5) \approx 1.758$$

$x_k = 1.3,\ \hat{\epsilon} = -0.5$	current state and predicted noise
$\alpha_k = 0.8,\ \sigma_k = 0.6$	current signal and noise scales
$\alpha_j = \sqrt{0.91},\ \sigma_j = 0.3$	scales at selected cleaner level j
$\hat{x}_0, x_j$	clean estimate and destination state



DDIM Eq.12; course example · Song, Meng & Ermon (2021)

EXERCISE

Reverse sampling

Transfer: check deterministic sampling at the endpoint

Find y. Can a new seed change a full sample?

$$\hat{x}_0 = (x_k - \sigma_k \hat{\epsilon})/\alpha_k, \qquad y = \alpha_0 \hat{x}_0 + \sigma_0 \hat{\epsilon}$$

$x_k = 0.9,\ \hat{\epsilon} = 0.5$	fixed input and prediction
$\alpha_k = 0.8,\ \sigma_k = 0.6$	current scales
$\alpha_0 = 1,\ \sigma_0 = 0$	endpoint scales
$\hat{x}_0$	clean estimate

DDIM Eq.12; course exercise · Song, Meng & Ermon (2021)

RESULT

Reverse sampling

A generalized step allocates fresh randomness

Allocate the destination noise variance.

$$x_j = \alpha_j \hat{x}_0 + \sqrt{\sigma_j^2 - \rho_{k,j}^2}\,\hat{\epsilon} + \rho_{k,j} z$$

x_j, α_j, σ_j	state, signal scale and noise scale at j < k
$z \sim \mathcal{N}(0, I)$	independent fresh draw
$0 \le \rho_{k,j} \le \sigma_j$	fresh noise scale
$\hat{x}_0, \hat{\epsilon}$	fixed clean and noise estimates

DDIM §4.1, Eq.12; §4.2 and Appendix C.1 · Song, Meng & Ermon (2021)

MODEL

Reverse sampling

The eta parameter selects the stochasticity

Eta controls the amount of fresh noise added by a sampling step.

$$\rho_{k,j} = \eta \sqrt{\frac{1-\alpha_j^2}{1-\alpha_k^2}\left(1-\frac{\alpha_k^2}{\alpha_j^2}\right)}$$

$\rho_{k,j}$	fresh noise standard deviation
$j < k,\ 0 \le \eta \le 1$	destination/current levels and noise scale
$0 < \alpha_k < \alpha_j \le 1$	current and destination signal amplitudes

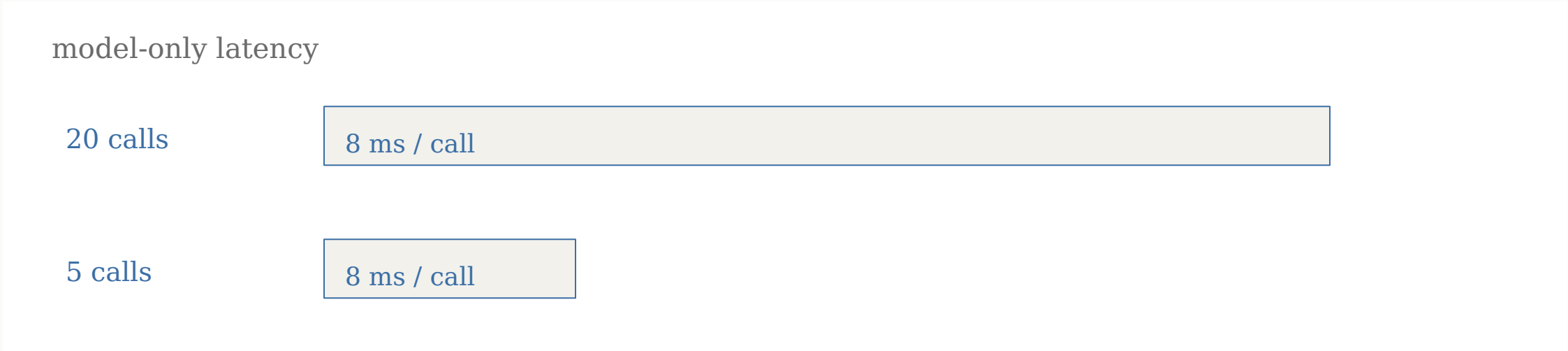
DDIM Eq.16; §4.1 and Appendix C.1 · Song, Meng & Ermon (2021)

EXERCISE

Reverse sampling

A step budget trades computation for approximation

Assumed cost: 8 ms per network evaluation.
Compare 20 evaluations with 5.
Which claims require an experiment?



Diffusion Policy §3.4; course exercise · Chi et al. (2023/2024)

DEFINITION

Conditioning

The action distribution depends on observations

The conditional score is the input gradient of a log density with the observation held fixed.

$$s_k(x \mid o) = \nabla_x \log p_k(x \mid o)$$

- o fixed observation context
- x noisy action column vector
- $p_k(x \mid o) > 0$ density differentiable in x at the point considered

Diffusion Policy §2.3 · Chi et al. (2023/2024)

MODEL

Conditioning

The denoiser has three inputs and one matching output

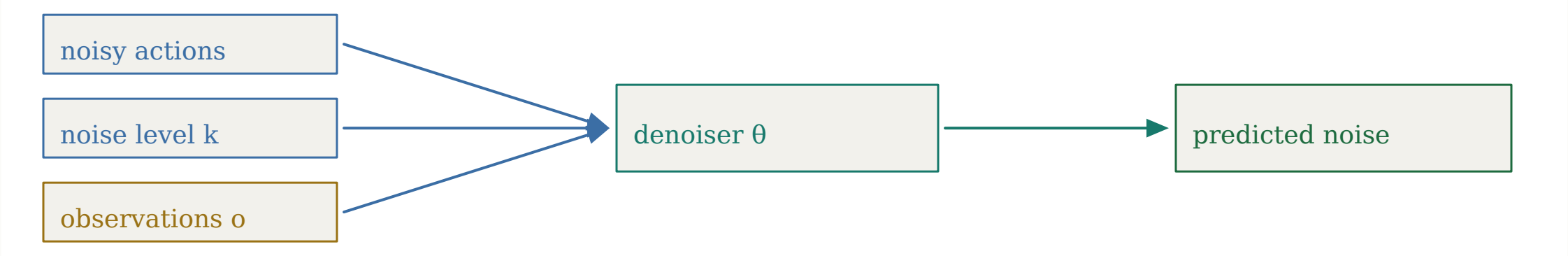
The denoiser maps noisy actions, a noise level and fixed observations to a same-shaped noise prediction.

$$\epsilon_\theta : (x_k, k, o) \longmapsto \hat{\epsilon} \in \mathbb{R}^d$$

$x_k \in \mathbb{R}^d$ noisy action sequence

k, o noise level and encoded observation

$\theta, \hat{\epsilon}$ network weights and predicted d-component noise



Diffusion Policy §§2.3,3.1–3.2 · Chi et al. (2023/2024)

DEFINITION

Conditioning

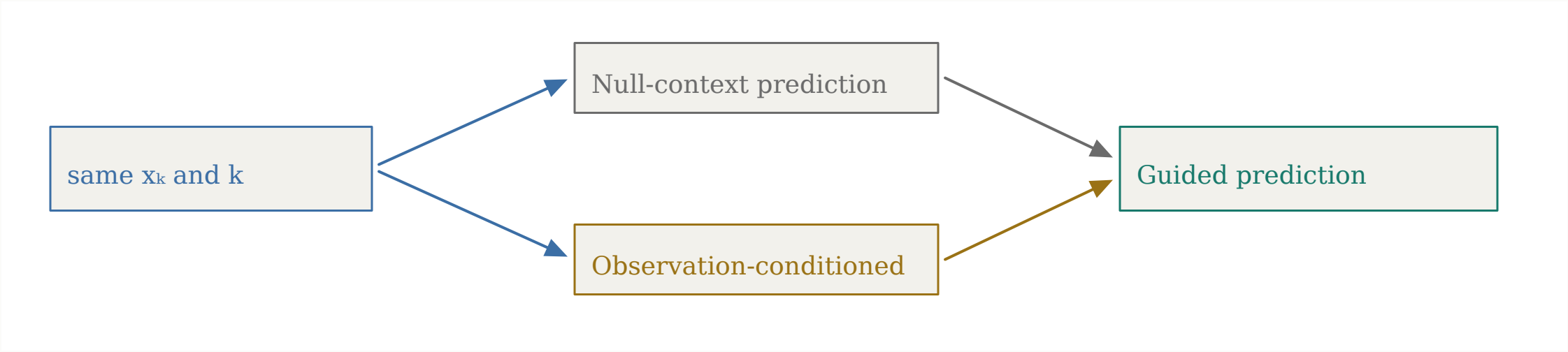
Classifier-free guidance combines two predictions

Classifier-free guidance combines conditional and unconditional denoiser predictions to control the effect of a condition.

$$\hat{\epsilon}_g = \hat{\epsilon}_u + w(\hat{\epsilon}_c - \hat{\epsilon}_u)$$

$\hat{\epsilon}_c, \hat{\epsilon}_u$ conditional and null-context predictions

w guidance scale; same x and k



Classifier-Free Guidance §3.2, Eq.6, Algorithms 1–2; w here = 1 + paper w · Ho & Salimans (2022)

INTERPRETATION

Conditioning

Guidance changes the fixed-level density direction

Use exact scores at one fixed noise level.

$$s_g = (1-w)s_u + w s_c = \nabla_x \log\left[p_k(x)^{1-w} p_k(x \mid o)^w\right]$$

$p_k(x), p_k(x \mid o)$ positive noisy densities without/with fixed observation

s_u, s_c, s_g unconditional, conditional and guided scores

w guidance scale

Classifier-Free Guidance §3.2; convention bridge · Ho & Salimans (2022)

Conditioning

Guidance changes the fixed-level density direction

Convert the linear prediction rule into scores

Both predictions use the same noise scale $\sigma_k > 0$.

1. $\hat{\epsilon}_g = (1 - w)\hat{\epsilon}_u + w\hat{\epsilon}_c$.
2. $s_g = -\hat{\epsilon}_g/\sigma_k = (1 - w)s_u + ws_c$.
3. For exact scores, $s_g = \nabla_x[(1 - w)\log p_k(x) + w \log p_k(x \mid o)]$.

Classifier-Free Guidance §3.2; convention bridge · Ho & Salimans (2022)

Conditioning

Guidance changes the fixed-level density direction

Combine logarithms with the necessary qualification

Positive densities at fixed k; a normalized interpretation requires finite integral of the product.

1. $s_g = \nabla_x \log[p_k(x)^{1-w} p_k(x \mid o)^w]$.
2. This is a fixed-noise-level identity. It does not alone identify the final distribution of a finite guided sampler.
3. The practical scale trades conditioning strength against diversity and artifacts.

Classifier-Free Guidance §3.2; convention bridge · Ho & Salimans (2022)

EXERCISE

Conditioning

Transfer: calculate and interpret guidance

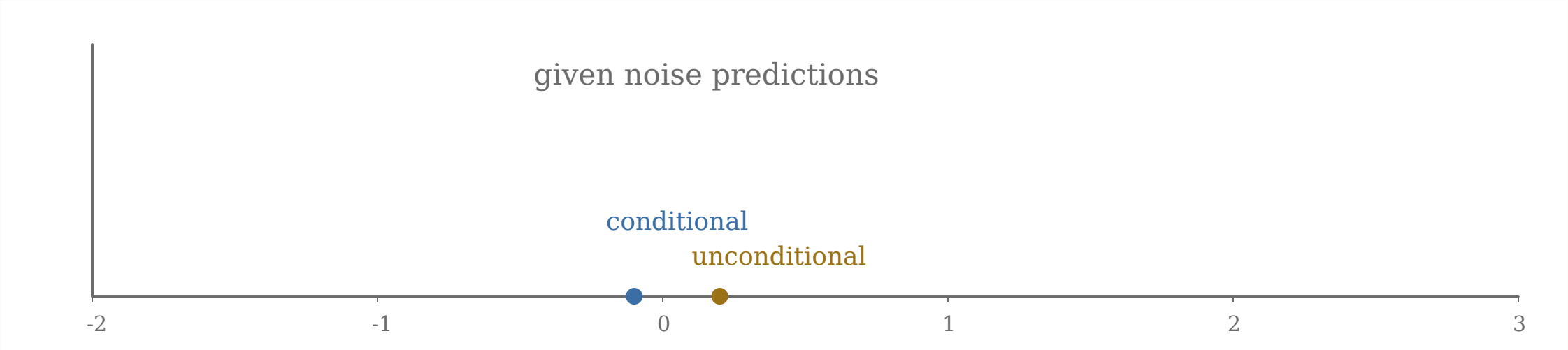
Compute guided noise and score for each w.

$$\hat{\epsilon}_g = \hat{\epsilon}_u + w(\hat{\epsilon}_c - \hat{\epsilon}_u), \qquad s_g = -\hat{\epsilon}_g / \sigma_k$$

$\hat{\epsilon}_u = 0.2,\ \hat{\epsilon}_c = -0.1$ null-context / conditional noise

$\sigma_k = 0.5$ shared noise standard deviation

$w = 0, 1, 2$ guidance scales to compare



Classifier-Free Guidance §3.2; course exercise · Ho & Salimans (2022)

MODEL

Diffusion Policy

An action chunk is the vector being generated

An action chunk is a sequence of future robot commands predicted together.

$$A_r^{(0)} = \begin{bmatrix} a_r^T \\ \vdots \\ a_{r+H_p-1}^T \end{bmatrix} \in \mathbb{R}^{H_p \times d_a}$$

- r physical robot time
- $a_r \in \mathbb{R}^{d_a}$ one column-vector action
- H_p, d_a prediction horizon and action dimension
- (0) clean diffusion level

Diffusion Policy §2.3 · Chi et al. (2023/2024)

EXERCISE

Diffusion Policy

Three horizons serve different purposes

Count observations, predicted scalars, and executed steps.

$$H_o = 2, \quad H_p = 4, \quad H_a = 2, \quad d_a = 2$$

- H_o recent observations supplied
- H_p future actions predicted
- H_a actions executed before replanning
- d_a scalar components of one action

Diffusion Policy §2.3; course exercise · Chi et al. (2023/2024)

DEFINITION

Diffusion Policy

Train on demonstrated chunks

Squared Frobenius norm sums squared entries.

$$L(\theta) = \mathbb{E}\|\epsilon_\theta(A_r^{(k)}, k, o_r) - \epsilon\|_F^2$$

$A_r^{(k)} = \alpha_k A_r^{(0)} + \sigma_k \epsilon$ noisy/clean chunks at time r

α_k, σ_k signal/noise scales

o_r, k observations, noise level

$\epsilon, \epsilon_\theta$ Gaussian noise and matching prediction

$\mathbb{E}, \theta$ avg over data, levels, noise; weights

Diffusion Policy §2.3, Eq.5; DDPM Eq.14 · Ho, Jain & Abbeel (2020) · Chi et al. (2023/2024)

EXERCISE

Diffusion Policy

A receding horizon closes the feedback loop

The robot predicts a chunk of actions and executes a prefix. It then observes a change in the scene. What can it do with the unused suffix?

Diffusion Policy §2.3 · Chi et al. (2023/2024)

EXAMPLE

Diffusion Policy

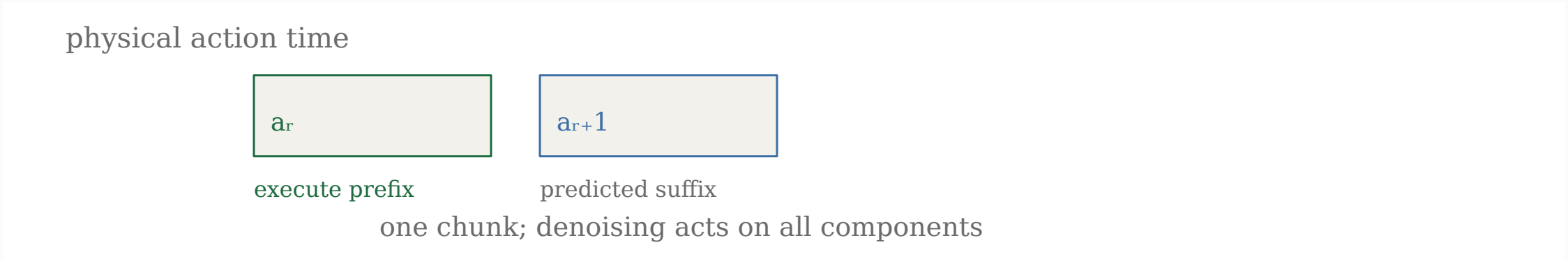
Recover and execute a two-action chunk

EXAMPLE

Recover both; execute the first.

$$x_1 = \begin{bmatrix} 1.3 \\ -0.2 \end{bmatrix}, \quad \hat{\epsilon} = \begin{bmatrix} -0.5 \\ 1 \end{bmatrix}, \quad \alpha_1 = 0.8,\ \sigma_1 = 0.6$$

$x_1, \hat{\epsilon}$	noisy chunk; predicted noise
α_1, σ_1	signal and noise scales
$x = (x_1 - \sigma_1\hat{\epsilon})/\alpha_1$	clean normalized chunk
$a = 0.4 + 0.1x$	each target in m



Diffusion Policy §§2.3,A.1; course example · Chi et al. (2023/2024)

EXAMPLE

Diffusion Policy

Integrated example: the final denoising step

EXAMPLE

$$\hat{x}_0 = \frac{x_1 - 0.6\hat{\epsilon}}{0.8} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$x_1 = (1.3, -0.2)^T$ noisy chunk at level one

$\hat{\epsilon} = (-0.5, 1)^T$ predicted noise

$0.8,\ 0.6$ level-one signal coefficient and noise standard deviation

$\hat{x}_0$ estimated normalized clean chunk

DDPM Eq.11; DDIM Eq.12; course example · Ho, Jain & Abbeel (2020) · Song, Meng & Ermon (2021)

EXAMPLE

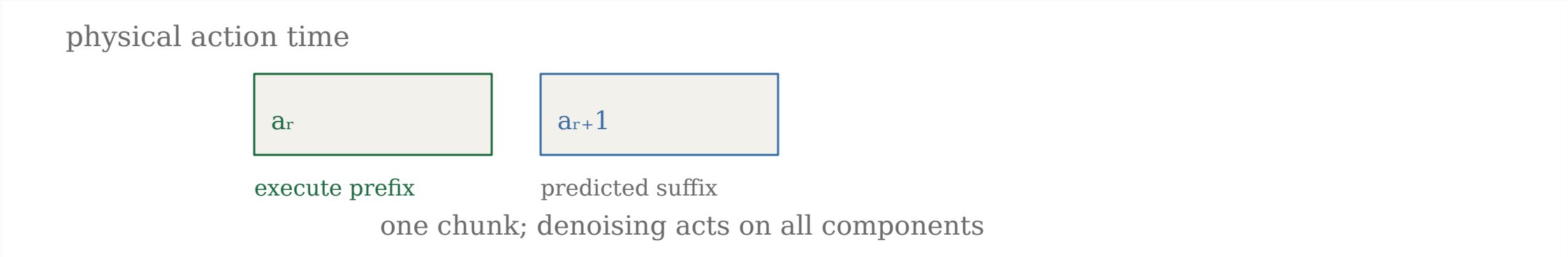
Diffusion Policy

Integrated example: restore units and execute a prefix

EXAMPLE

$$a = 0.4 + 0.1x, \quad x = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \quad \Rightarrow \quad A_r^{(0)} = \begin{bmatrix} 0.6 \\ 0.3 \end{bmatrix} \text{m}$$

- x dimensionless normalized target coordinates
- a target position; offset 0.4 m, scale 0.1 m per coordinate unit
- $A_r^{(0)}$ two clean targets at physical time r
- $H_p = 2, H_a = 1$ predict two; execute one



Diffusion Policy §2.3; course example · Chi et al. (2023/2024)

EXERCISE

Diffusion Policy

Transfer: change execution without changing the sampler

Targets: 0.6 m, then 0.3 m. H_a counts actions executed before replanning. Compare $H_a = 1$ and $H_a = 2$ when the obstacle moves after the first action.

Diffusion Policy §2.3; course exercise · Chi et al. (2023/2024)

INTERPRETATION

Diffusion Policy

ACT and diffusion share a chunked control interface

Action Chunking with Transformers (ACT) and diffusion policies both predict observation-conditioned action sequences.

	ACT	Diffusion
Training	reconstruct + regularize latent	noise prediction
Generation	test time: fix latent z = 0	iterative, random initialization
Output	action chunk	action chunk

ACT §§IV-B–IV-C, Fig.3, Algorithms 1–2; Diffusion Policy §§2–3 · Chi et al. (2023/2024) · Zhao et al. (2023)

EXERCISE

Diffusion Policy

A low denoising loss is not a deployment certificate

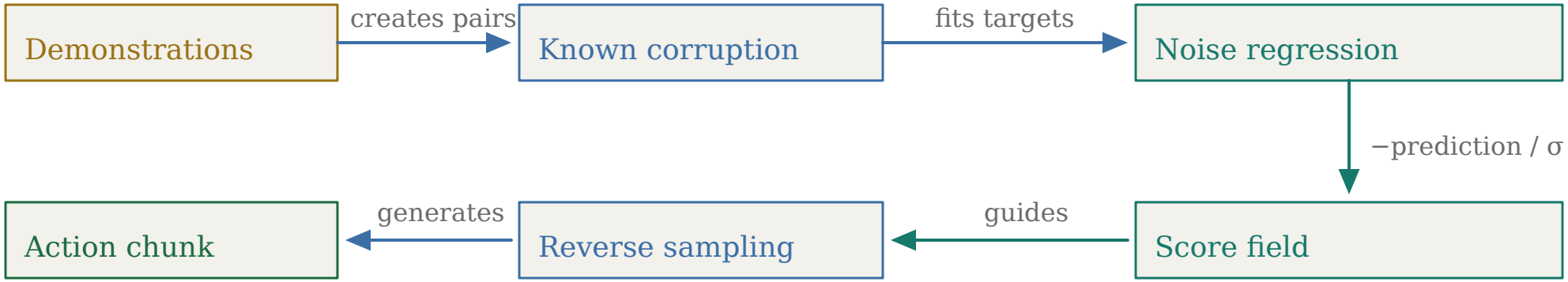
Training metric: noise-prediction error.
Deployment question: does the closed-loop task succeed?
Give two missing checks.

Diffusion Policy §§5,9; course exercise · Chi et al. (2023/2024)

RECAP

Diffusion and Diffusion Policy

The complete mathematical chain



DDPM §§2–3; Diffusion Policy §2.3 · Ho, Jain & Abbeel (2020) · Chi et al. (2023/2024)

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